SPM Derivation. (FDM)

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I>0 → discharge
I<0 → charge
→ scott.

$$\bar{r} = \frac{r}{R_s} \rightarrow r = R_s \cdot \bar{r} \qquad \partial r = R_s \cdot \partial \bar{r}$$

$$\frac{\partial}{\partial \bar{r}} C_s^{\pm}(i\Delta\bar{r}, t) \cong \frac{C_{s,i+1}^{\pm} - C_{s,i-1}^{\pm}}{2\Delta\bar{r}} \quad (\text{1st order central difference})$$

$$\frac{\partial^2}{\partial \bar{r}^2} C_s^{\pm}(i\Delta\bar{r}, t) = \frac{\frac{C_{s,i+1}^{\pm} - C_{s,i}^{\pm}}{\Delta\bar{r}} - \frac{C_{s,i}^{\pm} - C_{s,i-1}^{\pm}}{\Delta\bar{r}}}{\Delta\bar{r}} = \frac{C_{s,i+1}^{\pm} - 2C_{s,i}^{\pm} + C_{s,i-1}^{\pm}}{\Delta\bar{r}^2}$$

(2nd order central difference)

} FDM

$$\frac{\partial C_s^{\pm}}{\partial t}(\bar{r}, t) = D_s^{\pm}\left[\frac{2}{(R_s^{\pm})^2 \bar{r}} \cdot \frac{\partial C_s^{\pm}}{\partial \bar{r}}(\bar{r}, t) + \frac{1}{(R_s^{\pm})^2} \cdot \frac{\partial^2 C_s^{\pm}}{\partial \bar{r}^2}(\bar{r}, t)\right]$$

$$\frac{\partial C_s^{\pm}}{\partial \bar{r}}(0, t) = 0$$

$$\frac{\partial C_s^{\pm}}{\partial \bar{r}}(1^{\pm}, t) = \pm \frac{1}{D_s^{\pm} F a^{\pm} A^{\pm} L^{\pm}} I(t)$$

$$C_s^{\pm}(\bar{r}, t_0) = C_{s_0}^{\pm}(\bar{r})$$

} SPM

FDM for Conditions

$$\frac{\partial}{\partial \bar{r}} C_s^{\pm}(0^+, t) = \frac{-C_{s,2}^{\pm} + 4C_{s,1}^{\pm} - 3C_{s,0}^{\pm}}{2\Delta \bar{r} R_s^{\pm}}$$

(2nd order forward finite difference)

$$\frac{\partial}{\partial \bar{r}} C_s^{\pm}(1^{\pm}, t) = \frac{3C_{s,N}^{\pm} - 4C_{s,N-1}^{\pm} + C_{s,N-2}^{\pm}}{2\Delta \bar{r} R_s^{\pm}}$$

(2nd order ⊕ backward finite difference).

Check.

- 2nd order. Central difference
- 2nd order fwd finite difference
- 2nd order bkwd finite difference

$$\dot{C}_{s,i}^{+} = D_s^{+}\left[\frac{2}{(R_s^{+})^2 \bar{r}} \cdot \frac{\partial C_s^{+}}{\partial \bar{r}}(\bar{r},t) + \frac{1}{(R_s^{+})^2}\frac{\partial^2 C_s^{+}}{\partial \bar{r}^2}(\bar{r},t)\right]$$

$$= D_s^{+}\left[\frac{2}{(R_s^{+})^2 i\,\Delta\bar{r}}\left(\frac{C_{s,i+1}^{+} - C_{s,i-1}^{+}}{2\Delta\bar{r}}\right) + \frac{1}{(R_s^{+})^2}\left(\frac{C_{s,i+1}^{+} - 2C_{s,i}^{+} + C_{s,i-1}^{+}}{\Delta\bar{r}^2}\right)\right]$$

$$= \frac{D_s^{+}}{(R_s^{+})^2 (\Delta\bar{r})^2}\left[\frac{1}{i}\left(C_{s,i+1}^{+} - C_{s,i-1}^{+}\right) + \left(C_{s,i+1}^{+} - 2C_{s,i}^{+} + C_{s,i-1}^{+}\right)\right]$$

F

$$= \frac{D_s^{+}}{(R_s^{+})^2 (\Delta\bar{r})^2}\left[\frac{i-1}{i} C_{s,i-1}^{+} - 2C_{s,i}^{+} + \frac{i+1}{i} C_{s,i+1}^{+}\right]$$

$\Rightarrow \alpha^{+}$

$$\dot{C}_{s,1}^{+} = \alpha^{+}\left(0 \cdot C_{s,0}^{+} - 2C_{s,1}^{+} + 2 \cdot C_{s,2}^{+}\right)$$

$$\dot{C}_{s,2}^{+} = \alpha^{+}\left(\frac{1}{2} \cdot C_{s,1}^{+} - 2C_{s,2}^{+} + \frac{3}{2} C_{s,3}^{+}\right)$$

$$\vdots$$

$$\dot{C}_{s,N-1}^{+} = \alpha^{+}\left(\frac{N-2}{N-1} C_{s,N-2}^{+} - 2C_{s,N-1}^{+} + \frac{N}{N-1} C_{s,N}^{+}\right)$$

BC's.

$$\frac{\partial C_s}{\partial \bar{r}}(0,t) = \frac{-C_{s,2}^{+} + 4C_{s,1}^{+} - 3C_{s,0}^{+}}{2\Delta\bar{r} R_s^{+}} = 0.$$

$$= -C_{s,2}^{+} + 4C_{s,1}^{+} - 3C_{s,0}^{+} = 0.$$

$$\frac{\partial C_s^{+}}{\partial \bar{r}}(1,t) = \frac{3C_{s,N}^{+} - 4C_{s,N-1}^{+} + C_{s,N-2}^{+}}{2\Delta\bar{r} \cdot R_s^{+}} = \frac{I(t)}{D_s^{+} F \alpha^{+} A^{+} L^{+}}$$

$$3C_{s,N}^{+} - 4C_{s,N-1}^{+} + C_{s,N-2}^{+} = \frac{2\Delta\bar{r} R_s^{+}}{D_s^{+} F \alpha^{+} A^{+} L^{+}} I(t).$$

$$\underbrace{\begin{bmatrix} \dot{C}_{s,1} \\ \dot{C}_{s,2} \\ \vdots \\ \dot{C}_{s,N-1} \end{bmatrix}}_{\dot{C}_s^+} = \underbrace{\begin{bmatrix} -2\alpha^+ & 2\alpha^+ & & \\ \frac{1}{2}\alpha^+ & -2\alpha^+ & \frac{3}{2}\alpha^+ & \\ 0 & \frac{2}{3}\alpha^+ & -2\alpha^+ & \frac{4}{3}\alpha^+ \\ & & & \ddots \\ & & & -2\alpha^+ \end{bmatrix}}_{M_1^+} \underbrace{\begin{bmatrix} C_{s,1}^+ \\ C_{s,2}^+ \\ \vdots \\ C_{s,N-1}^+ \end{bmatrix}}_{C_s^+} + \underbrace{\begin{bmatrix} 0 & 0 \\ \vdots & \vdots \\ 0 & 0 \\ 0 & \frac{N}{N-1}\alpha^+ \end{bmatrix}}_{M_2^+} \underbrace{\begin{bmatrix} C_{s,0}^+ \\ C_{s,N}^+ \end{bmatrix}}_{C_{s,2}^+}$$

— dynamics

$$\underbrace{\begin{bmatrix} 4 & -1 & 0 & \cdots & 0 \\ 0 & \cdots & & 1 & -4 \end{bmatrix}}_{N_1^+} \underbrace{\begin{bmatrix} C_{s,1}^+ \\ \vdots \\ C_{s,N-2}^+ \\ C_{s,N-1}^+ \end{bmatrix}}_{C_s^+} + \underbrace{\begin{bmatrix} -3 & 0 \\ 0 & 3 \end{bmatrix}}_{N_2^+} \underbrace{\begin{bmatrix} C_{s,0}^+ \\ C_{s,N}^+ \end{bmatrix}}_{C_{s,2}^+} = \underbrace{\begin{bmatrix} 0 \\ \frac{2\Delta r R_s^+}{D_s^+ F a^+ A^+ L^+} \end{bmatrix}}_{N_3^+} I(t)$$

— B.C.

$$= \begin{bmatrix} 0 \\ \frac{2\Delta r}{D_s} \end{bmatrix}$$

$$\alpha = \frac{D_s}{(\Delta r)^2}$$

$$\dot{C}_s^+ = M_1^+ C_s^+ + M_2^+ C_{s,2}^+$$

$$N_3^+ I(t) = N_1^+ C_s^+ + N_2^+ C_{s,2}^+$$

Solve for $C_{s,2}^+$ ⟹ output

$$C_{s,2}^+ = (N_2^+)^{-1} N_3^+ I(t) - (N_2^+)^{-1} N_1^+ C_s^+$$

plug into dynamics,

$$\dot{C}_s^+ = M_1^+ C_s^+ + M_2^+ \left((N_2^+)^{-1} N_3^+ I(t) - (N_2^+)^{-1} N_1^+ C_s^+ \right)$$

$$= \left(M_1^+ - M_2^+ (N_2^+)^{-1} N_1^+ \right) C_s^+ + M_2^+ (N_2^+)^{-1} N_3^+ I(t)$$

√ For Anode, switch sign of (N_3^+)